

Automating Quantified Multimodal Logics in HOL

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My very first Region Connection Calculus Prover

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Simple Type Theory / HOL

Simple Type Theory / HOL

- ▶ simple types $\alpha, \beta ::= \iota \mid o \mid \alpha \rightarrow \beta$ (additional base types μ_i)
- ▶ simple type theory / HOL defined by

$$s, t ::= p_\alpha \mid X_\alpha \mid (\lambda X_\alpha. s_\beta)_{\alpha \rightarrow \beta} \mid (s_{\alpha \rightarrow \beta} t_\alpha)_\beta \mid (\neg_{o \rightarrow o} s_o)_o \mid \\ (s_o \vee_{o \rightarrow o \rightarrow o} t_o)_o \mid (\Pi_{(\alpha \rightarrow o) \rightarrow o} t_{\alpha \rightarrow o})_o \mid (s_\alpha =_{\alpha \rightarrow \alpha \rightarrow o} t_\alpha)_o$$

- ▶ semantics well understood [Henkin50, Andrews72a/b, Benz MüllerEtAl04]
 - Henkin semantics
- ▶ base logic of many (interactive) proof assistants:
Isabelle/HOL, HOL, HOL-light, PVS, OMEGA, ...
- ▶ (too) few ATPs so far



THFTPTP

(EU grant THFTPTP – PIIF-GA-2008-219982)

Thanks to hard working Geoff Sutcliffe

THFTPTP – Progress in ATP for HOL

- ▶ THF syntax for HOL
- ▶ library for HOL (> 1275 problems)
- ▶ tools for HOL
(parser, type checker, pretty printer, ...)
- ▶ integrated HOL ATPs: IsabelleP, TPS, LEO-II
- ▶ integrated HOL model generator: IsabelleM
- ▶ SystemOnTPTP online interface

THFTPTP – Progress in ATP for HOL

ALG	50 - higher-order abstract syntax
GRA	93 - Ramsey numbers (several open)
LCL	61 - modal logic
NUM	205 - Landau's Grundlagen
PUZ	5 - puzzles
SET/SEU	749 - set theory, dependently typed set theory, binary relations
SWV	37 - security, access control logic
SYN/SYO	59 - simple test problems

	ALG	GRA	LCL	NUM	PUZ	SE?	SWV	SY?	Total	Unique
Problems	50	93	61	221	5	749	37	59	1275	
THM/UNS	50	25	51	221	5	746	25	47	1170	
CSA/SAT	0	0	10	0	0	3	5	11	29	
LEO-II 0.99a	34	0	48	181	3	401	19	42	725	127
IsabelleP 2008	0	0	0	197	5	361	1	30	594	74
TPS 3.0	10	0	40	150	3	285	9	35	532	6
Any	32	0	50	203	5	490	20	52	843	207
All	0	0	0	134	2	214	0	22	372	
None	18	93	12	18	0	259	17	15	432	
IsabelleM 2008	0	0	1	0	0	0	0	8	9	



LEO-II

(EPSRC grant EP/D070511/1 at Cambridge University)

Thanks to Larry Paulson

LEO-II

UNIVERSITY OF CAMBRIDGE
UNIVERSITÄT DES SAARLANDES

An Effective Higher-Order Theorem Prover

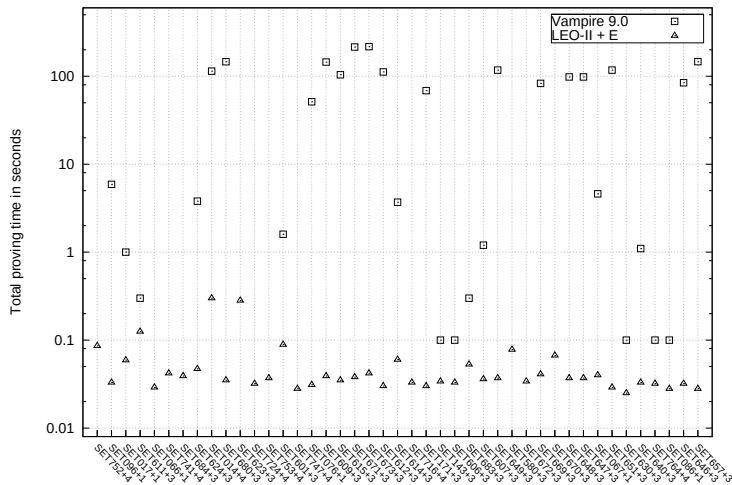


LEO-II employs FO-ATPs:

E, Spass, Vampire

<http://www.ags.uni-sb.de/~leo>

Observation: HOL can outperform FOL in some domains





Quantified Multimodal Logics as HOL Fragments

(jww Larry Paulson)

Quantified Multimodal Logics

- ▶ Quantified Multimodal Logics (QML) defined by

$$s, t ::= P \mid k(X^1, \dots, X^n) \mid \neg s \mid s \vee t \mid \forall X. s \mid \forall P. s \mid \Box_r s$$

- ▶ Kripke style semantics [Fitting05]

- ▶ ...
- ▶ different notions of models:
 - $\mathbf{QK}_{\pi^-}$
 - $\mathbf{QK}_{\pi}$
 - $\mathbf{QK}_{\pi^+}$

Embedding QML in HOL

Simple, Straightforward Encoding

- ▶ base type ι : non-empty set of possible worlds
- ▶ base type μ : non-empty set of individuals
- ▶ (certain) terms of type $\iota \rightarrow o$: QML formulas

Local Definitions of Modal Operators

$$\neg_{(\iota \rightarrow o) \rightarrow (\iota \rightarrow o)} = \lambda \phi_{\iota \rightarrow o}. \lambda W_{\iota}. \neg \phi W$$

$$\vee_{(\iota \rightarrow o) \rightarrow (\iota \rightarrow o) \rightarrow (\iota \rightarrow o)} = \lambda \phi_{\iota \rightarrow o}. \lambda \psi_{\iota \rightarrow o}. \lambda W_{\iota}. \phi W \vee \psi W$$

$$\Box_{(\iota \rightarrow \iota \rightarrow o) \rightarrow (\iota \rightarrow o) \rightarrow (\iota \rightarrow o)} = \lambda R_{\iota \rightarrow \iota \rightarrow o}. \lambda \phi_{\iota \rightarrow o}. \lambda W_{\iota}. \forall V_{\iota}. \neg R W V \vee \phi V$$

$$\prod_{(\mu \rightarrow (\iota \rightarrow o)) \rightarrow (\iota \rightarrow o)}^{\mu} = \lambda \phi_{\mu \rightarrow (\iota \rightarrow o)}. \lambda W_{\iota}. \forall X_{\mu}. \phi X W$$

$$\prod_{((\iota \rightarrow o) \rightarrow (\iota \rightarrow o)) \rightarrow (\iota \rightarrow o)}^{\iota \rightarrow o} = \lambda \phi_{(\iota \rightarrow o) \rightarrow (\iota \rightarrow o)}. \lambda W_{\iota}. \forall P_{\iota \rightarrow o}. \phi P W$$

Definition of Validity

Definition of validity

$$\text{valid}_{(\iota \rightarrow o) \rightarrow o} = \lambda \phi_{\iota \rightarrow o}. \forall W_{\iota}. \phi W$$

Soundness and Completeness

[BenzmüllerPaulson2009-submitted]

Soundness If $\models^{\mathcal{HOL}} (\text{valid } s_{l \rightarrow o})$ then $\models^{\mathbf{QK}^\pi} s$.

Completeness If $\models^{\mathbf{QK}^\pi} s$ then $\models^{\mathcal{HOL}} (\text{valid } s_{l \rightarrow o})$.

Furthermore

Corollary I Embedding of propositional QML is sound and complete.

Corollary II Embedding of first-order QML is sound and complete.

Corollary III Embedding of propositional multimodal logics is sound and complete.

Epistemic Reasoning about QML: Correspondences

Correspondences between properties of accessibility relation and axioms

reflexive $= \lambda R. \forall S. R S S$	(1)	$\forall A. \Box_r A \supset A$	(11)
symmetric $= \lambda R. \forall S, T. ((R S T) \Rightarrow (R T S))$	(2)	$\forall A. A \supset \Box_r \Diamond_r A$	(12)
serial $= \lambda R. \forall S. \exists T. (R S T)$	(3)	$\forall A. \Box_r A \supset \Diamond_r A$	(13)
transitive $= \lambda R. \forall S, T, U. ((R S T) \wedge (R T U) \Rightarrow (R S U))$	(4)	$\forall A. \Box_r A \supset \Box_r \Box_r A$	(14)
euclidean $= \lambda R. \forall S, T, U. ((R S T) \wedge (R S U) \Rightarrow (R T U))$	(5)	$\forall A. \Diamond_r A \supset \Box_r \Diamond_r A$	(15)
partially_functional $= \lambda R. \forall S, T, U. ((R S T) \wedge (R S U) \Rightarrow T = U)$	(6)	$\forall A. \Diamond_r A \supset \Box_r A$	(16)
functional $= \lambda R. \forall S. \exists T. ((R S T) \wedge \forall U. ((R S U) \Rightarrow T = U))$	(7)	$\forall A. \Diamond_r A \Leftrightarrow \Box_r A$	(17)
weakly_dense $= \lambda R. \forall S, T. ((R S T) \Rightarrow \exists U. ((R, S U) \wedge (R U T)))$	(8)	$\forall A. \Box_r \Box_r A \supset \Box_r A$	(18)
weakly_connected = ...	(9)	$\forall A, B. \dots$	(19)
weakly_directed = ...	(10)	$\forall A. \Diamond_r \Box_r A \supset \Box_r \Diamond_r A$	(20)

Epistemic Reasoning about QML: Correspondences

Correspondences between properties of accessibility relation and axioms

► $(2) \Rightarrow (12)$

$$\forall R. (\text{symmetric } R) \Rightarrow (\text{valid } \forall A. A \supset \Box_r \Diamond_r A)$$

► $(2) \Leftarrow (12)$

$$\forall R. (\text{symmetric } R) \Leftarrow (\text{valid } \forall A. A \supset \Box_r \Diamond_r A)$$

Automating Reasoning about QML: Correspondences

Problem	IsabelleP	LEO-II	TPS
(1) $\Rightarrow$ (11)	THM(90.0)	THM(0.0)	THM(0.3)
(2) $\Rightarrow$ (12)	THM(113.7)	THM(0.0)	THM(0.2)
(3) $\Rightarrow$ (13)	THM(102.1)	THM(0.0)	THM(0.3)
(4) $\Rightarrow$ (14)	THM(123.2)	THM(0.0)	THM(8.1)
(5) $\Rightarrow$ (15)	THM(131.7)	THM(0.0)	THM(0.6)
(6) $\Rightarrow$ (16)	THM(125.8)	THM(0.0)	THM(1.1)
(7) $\Rightarrow$ (17)	THM(156.3)	THM(0.0)	–
(8) $\Rightarrow$ (18)	THM(105.1)	THM(0.0)	THM(1.1)
(9) $\Rightarrow$ (19)	THM(42.6)	THM(0.0)	THM(0.2)
(10) $\Rightarrow$ (20)	THM(178.1)	THM(0.0)	THM(0.2)
(1) $\Leftarrow$ (11)	THM(75.3)	THM(0.0)	THM(0.2)
(2) $\Leftarrow$ (12)	–	–	THM(0.5)
(3) $\Leftarrow$ (13)	THM(89.6)	THM(0.0)	THM(0.2)
(4) $\Leftarrow$ (14)	–	–	–
(5) $\Leftarrow$ (15)	–	–	–
(6) $\Leftarrow$ (16)	–	THM(6.2)	THM(47.9)
(7) $\Leftarrow$ (17)	–	–	THM(151.3)
(8) $\Leftarrow$ (18)	–	–	–
(9) $\Leftarrow$ (19)	–	–	–
(10) $\Leftarrow$ (20)	THM(85.1)	THM(0.1)	THM(148.5)

Automating Reasoning in QML: Wise Men Puzzle

Wise men puzzle Once upon a time, a king wanted to find the wisest out of his three wisest men. He arranged them in a circle and told them that he would put a white or a black spot on their foreheads and that one of the three spots would certainly be white. The three wise men could see and hear each other but, of course, they could not see their faces reflected anywhere. The king, then, asked to each of them to find out the color of his own spot. After a while, the wisest correctly answered that his spot was white.

Modeling in QML (extending [BaldoniPhD03])

- ▶ declare accessibility relation constants: $a_{l \rightarrow l \rightarrow o}, b_{l \rightarrow l \rightarrow o}, c_{l \rightarrow l \rightarrow o}, \text{fool}_{l \rightarrow l \rightarrow o}$
- ▶ declare 'white spot' function: $ws_{(l \rightarrow l \rightarrow o) \rightarrow l \rightarrow o}$
- ▶ common knowledge: at least one of the wise men has a white spot

$$\text{valid } \Box_{\text{fool}} (ws\ a) \vee (ws\ b) \vee (ws\ c)$$

- ▶ common knowledge: if someone has a white spot the others can see this

$$\forall X, Y. (\text{cond } X\ Y) \Rightarrow (\text{valid } \Box_{\text{fool}} (ws\ X) \Rightarrow \Box_Y (ws\ X))$$

$$\text{cond} = \lambda X, Y. (X = a \wedge Y = b) \vee (X = a \wedge Y = c) \vee \dots \vee (X = c \wedge Y = b)$$

Automating Reasoning in QML: Wise Men Puzzle

Modeling cont'd

- ▶ common knowledge: if someone has not a white spot the others can see this

$$\forall X, Y. (\text{cond } X \ Y) \Rightarrow (\text{valid } \Box_{\text{fool}} \neg (ws \ X) \Rightarrow \Box_Y \neg (ws \ X))$$

- ▶ T, K, I axioms for common knowledge

$$\text{valid } \forall A. \Box_{\text{fool}} A \Rightarrow A$$

$$\text{valid } \forall A. \Box_{\text{fool}} A \Rightarrow \Box_{\text{fool}} \Box_{\text{fool}} A$$

$$\forall X. \text{valid } \forall A. \Box_{\text{fool}} A \Rightarrow \Box_X A$$

- ▶ if someone knows A then the others know that he knows A

$$\forall X, Y. (\text{cond } X \ Y) \Rightarrow (\text{valid } \forall A. (\Box_X A \Rightarrow \Box_Y \Box_X A))$$

- ▶ if someone does not know A then the others know that he does not know A

$$\forall X, Y. (\text{cond } X \ Y) \Rightarrow (\text{valid } \forall A. (\neg \Box_X A \Rightarrow \Box_Y \neg \Box_X A))$$

- ▶ a and b do not know that they have a white spot

$$\text{valid } \neg \Box_a (ws \ a) \qquad \text{valid } \neg \Box_b (ws \ b)$$

- ▶ prove that c does know he has a white spot:

$$\text{valid } \Box_c (ws \ c)$$

Last Night: My first Region Connection Calculus Prover

RCC-8 Relation	Topological Constraints
DC(a, b)	$\bar{a} \cap \bar{b} = \emptyset$
EC(a, b)	$i(\bar{a}) \cap i(\bar{b}) = \emptyset, \bar{a} \cap \bar{b} \neq \emptyset$
PO(a, b)	$i(\bar{a}) \cap i(\bar{b}) \neq \emptyset, \bar{a} \not\subseteq \bar{b}, \bar{b} \not\subseteq \bar{a}$
TPP(a, b)	$\bar{a} \subset \bar{b}, \bar{a} \not\subseteq i(\bar{b})$
TPP ⁻¹ (a, b)	$\bar{b} \subset \bar{a}, \bar{b} \not\subseteq i(\bar{a})$
NTPP(a, b)	$\bar{a} \subset i(\bar{b})$
NTPP ⁻¹ (a, b)	$\bar{b} \subset i(\bar{a})$
EQ(a, b)	$\bar{a} = \bar{b}$

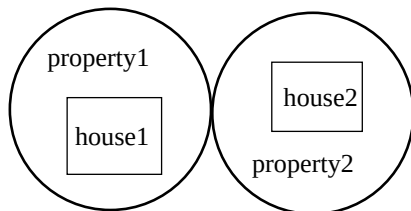
$$ec = \lambda A_{\iota \rightarrow o}, B_{\iota \rightarrow o}. i(\bar{A}) \cap i(\bar{B}) = \emptyset \wedge \bar{A} \cap \bar{B} \neq \emptyset$$

$$ec' = \lambda A_{\iota \rightarrow o}, B_{\iota \rightarrow o}. \neg(\exists U. U \in i(\bar{A}) \cap i(\bar{B})) \wedge (\exists U. U \in \bar{A} \cap \bar{B})$$

Last Night: My first Region Connection Calculus Prover

$\circ$	DC	EC	PO	TPP	NTPP	TPP ⁻¹	NTPP ⁻¹	EQ
DC	*	DC,EC PO,TPP NTPP	DC,EC PO,TPP NTPP	DC,EC PO,TPP NTPP	DC,EC PO,TPP NTPP	DC	DC	DC
EC	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	DC,EC PO,TPP TPP ⁻¹ ,EQ	DC,EC PO,TPP NTPP	EC,PO TPP NTPP	PO TPP NTPP	DC,EC	DC	EC
PO	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	*	PO TPP NTPP	PO TPP NTPP	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	PO
TPP	DC	DC,EC	DC,EC PO,TPP NTPP	TPP NTPP	NTPP	DC,EC PO,TPP TPP ⁻¹ ,EQ	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	TPP
NTPP	DC	DC	DC,EC PO,TPP NTPP	NTPP	NTPP	DC,EC PO,TPP NTPP	*	NTPP
TPP ⁻¹	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	EC,PO TPP ⁻¹ NTPP ⁻¹	PO TPP ⁻¹ NTPP ⁻¹	PO,EQ TPP TPP ⁻¹	PO TPP NTPP	TPP ⁻¹ NTPP ⁻¹	NTPP ⁻¹	TPP ⁻¹
NTPP ⁻¹	DC,EC PO,TPP ⁻¹ NTPP ⁻¹	PO TPP ⁻¹ NTPP ⁻¹	PO TPP ⁻¹ NTPP ⁻¹	PO TPP ⁻¹ NTPP ⁻¹	PO,TPP ⁻¹ TPP,NTPP NTPP ⁻¹ ,EQ	NTPP ⁻¹	NTPP ⁻¹	NTPP ⁻¹
EQ	DC	EC	PO	TPP	NTPP	TPP ⁻¹	NTPP ⁻¹	EQ

Last Night: My first Region Connection Calculus Prover



```
thf(a1,axiom, (ec @ property1 @ property2) ).  
thf(a2,axiom, (ntpp @ house1 @ property1) ).  
thf(a3,axiom, (ntpp @ house2 @ property2) ).  
thf(conj,conjecture, (dc @ house1 @ house2) ).
```

LEO-II proves this in 0.15 seconds

Summary and Outlook

- ▶ Reasoning in and about QML can be elegantly encoded in HOL
- ▶ Some interesting problems automatically solvable
- ▶ Quick and easy combination of specific logics
- ▶ Applicable to mechanization/automation of aspects of time and space?
 - ▶ via QML in HOL
 - ▶ directly in HOL
 - ▶ mixtures of both
- ▶ Exploration and verification of theories
- ▶ Rapid prototyping
- ▶ Challenge problems for HOL library / cross-comparison